RAI 811: ROBOT MECHANICS & CONTROL

Lecture 07:
Inverse Kinematics

Outline

- **Existence of a Solution**
- **Multiple Solutions**

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- **Existence of a Solution**
- Multiple Solutions

Existence of a Solution

- **Workspace:** is defined as the volume of space that the end-effector (gripper) can reach. For a solution to exist, the end-effector goal must be within the workspace.

- **Degrees of Freedom (DOF):** Degrees of freedom are the number of independent variable in a manipulator. For a solution to exist, the number of joints should be more than or equal to the number of independent Cartesian coordinates to be controlled at the end effector.

Existence of a Solution: Degrees of Freedom

A **planar manipulator** is a robot arm whose motion is restricted to a single plane (usually the **XY-plane**). All its joint axes are parallel (typically along the **z-axis**), so the robot moves in 2D space.

For a **3R planar manipulator**:

•All joints rotate about the **z-axis**

•Motion is restricted to the **XY-plane**

•The end-effector can:

Move in **x-direction**

Move in **y-direction**

Rotate about **z-axis**

It **cannot**:

Move along the z-axis

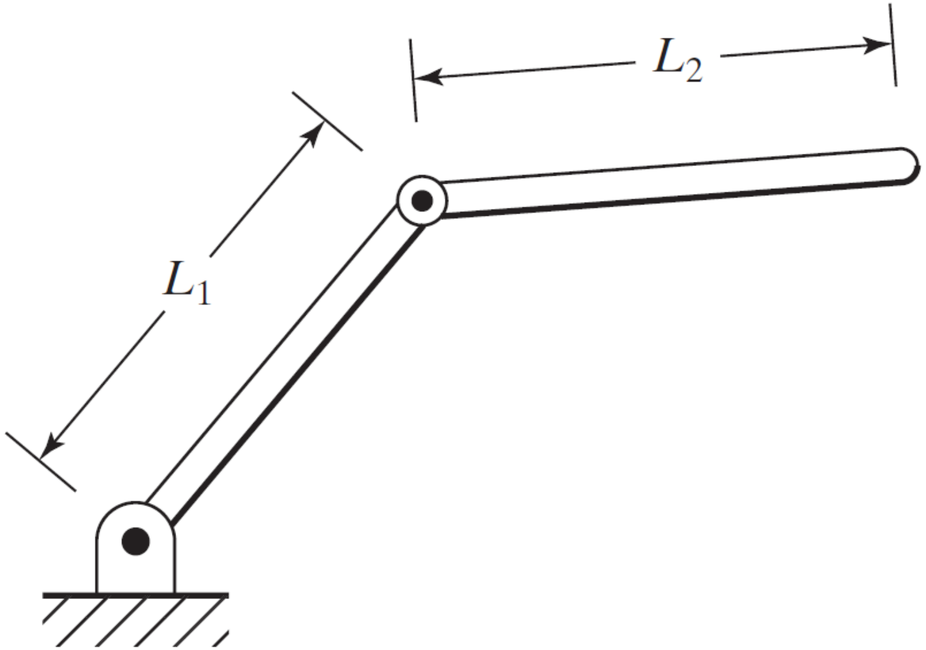
Rotate about the x-axis

Rotate about the y-axis

Existence of a Solution: Workspace

Example:

For the shown 2R planar manipulator, determine the conditions where an inverse Kinematics solution exists.



Existence of a Solution: Workspace

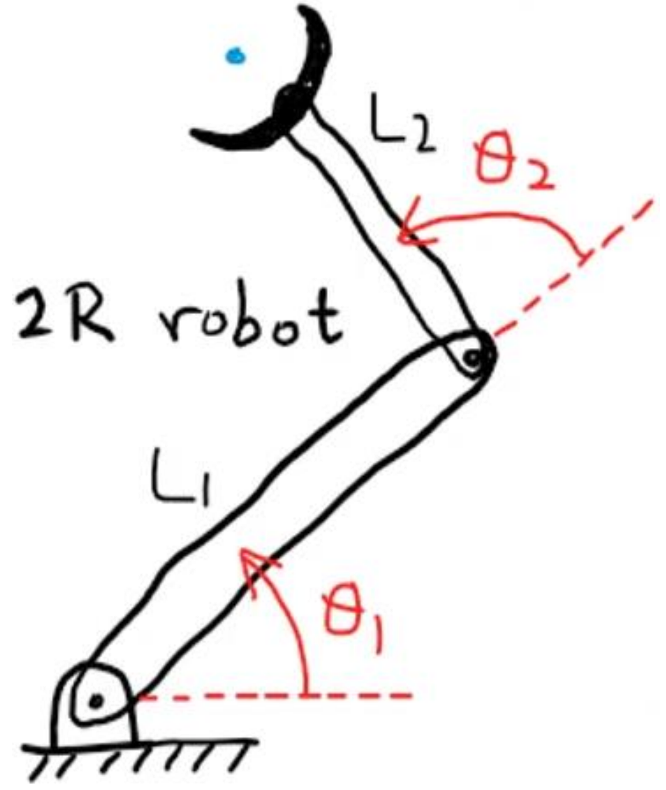
Example:

For the shown 2R planar manipulator, determine the conditions where an inverse Kinematics solution exists.

For an IK solution to exist, the workspace boundary is determined as follows:

- When the arm is fully stretched ($\theta_2 = 0°$), the outer workspace boundary is $(L_1 + L_2)$ from $\{0\}$ origin.
- When the arm is fully folded ($\theta_2 = 180°$), the inner workspace boundary is $(L_1 - L_2)$ from $\{0\}$ origin.

For a solution to exist, the destination point for the end-effector must be within the workspace area.

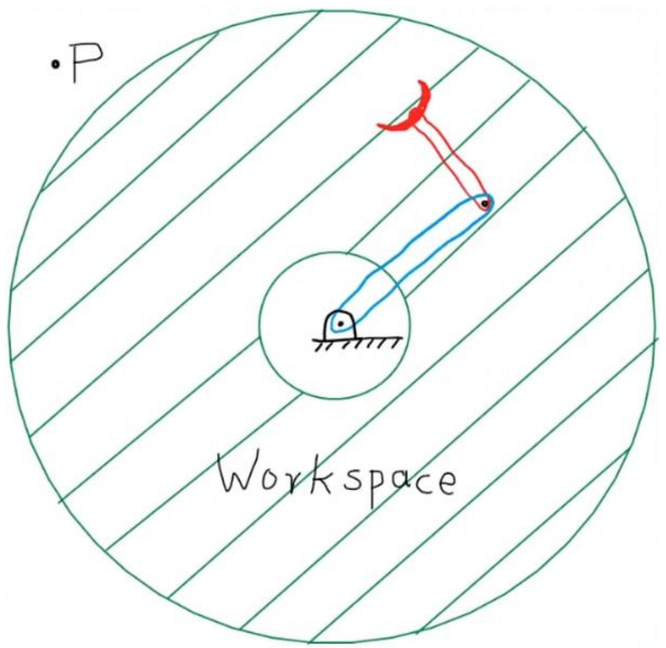


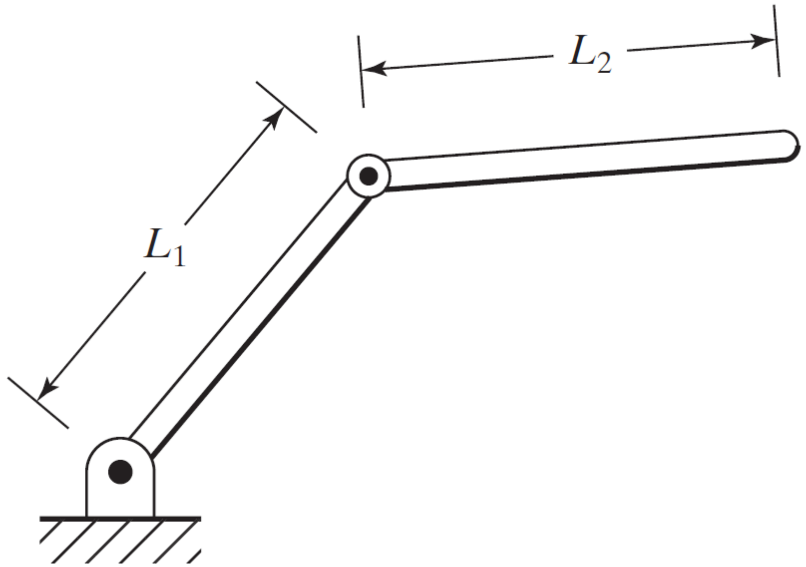
Existence of a Solution: Workspace

Example:

For the shown 2R planar manipulator, determine the conditions where an inverse Kinematics solution exists.

For a solution to exist, the destination point for the end-effector must be within the workspace area.





Existence of a Solution: Degrees of Freedom

Example:

For the 3R planar manipulator, the transformation matrix that relates frame {3} to the ground frame {0} is defined below. Can a solution exist if the desired transformation at frame {3} is of the following form, and why:

$$ {}^0_3T = \begin{bmatrix} c_{123} & -s_{123} & 0.0 & l_1c_1 + l_2c_{12} \\ s_{123} & c_{123} & 0.0 & l_1s_1 + l_2s_{12} \\ 0.0 & 0.0 & 1.0 & 0.0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^0_3T_{desired} = \begin{bmatrix} 0.866 & -0.5 & 0 & 6 \\ 0.5 & 0.866 & 0 & 5 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

(a)

$$ {}^0_3T_{desired} = \begin{bmatrix} 0.866 & 0 & 0.5 & 6 \\ 0 & 1 & 0 & 5 \\ -0.5 & 0 & 0.866 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

(b)

Existence of a Solution: Degrees of Freedom

Example:

For the 3R planar manipulator, the transformation matrix that relates frame {3} to the ground frame {0} is defined below. Can a solution exist if the desired transformation at frame {3} is of the following form, and why:

$$ {}^0_3T = \begin{bmatrix} c_{123} & -s_{123} & 0.0 & l_1c_1 + l_2c_{12} \\ s_{123} & c_{123} & 0.0 & l_1s_1 + l_2s_{12} \\ 0.0 & 0.0 & 1.0 & 0.0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

(a) A solution can exist, because the robot is 3 DoF and we can extract three unique equations from the transformation matrix (one in x, one in y, and one in R_z)

$$ {}^0_3T_{desired} = \begin{bmatrix} 0.866 & -0.5 & 0 & 6 \\ 0.5 & 0.866 & 0 & 5 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

Existence of a Solution: Degrees of Freedom

Example:

For the 3R planar manipulator, the transformation matrix that relates frame {3} to the ground frame {0} is defined below. Can a solution exist if the desired transformation at frame {3} is of the following form, and why:

(b) A solution does not exist, because the desired transformation matrix requires the robot to move in Cartesian coordinates along the z-axis and rotate about the y-axis, which the planar robot cannot do.

$$
{}^0_3T = \begin{bmatrix} c_{123} & -s_{123} & 0.0 & l_1c_1 + l_2c_{12} \\ s_{123} & c_{123} & 0.0 & l_1s_1 + l_2s_{12} \\ 0.0 & 0.0 & 1.0 & 0.0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

$$
{}^0_3T_{desired} = \begin{bmatrix} 0.866 & 0 & 0.5 & 6 \\ 0 & 1 & 0 & 5 \\ -0.5 & 0 & 0.866 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

Outline

- Existence of a Solution
- **Multiple Solutions**

Multiple Solutions

Multiple solutions exist if the robot can reach the desired goal in more than one configuration.

- **Finite number of solutions:** if the number of joints in the robot (DOF) is equal to the number of independent Cartesian coordinates that can be controlled within the robot's workspace. In 3D space, the maximum number of Cartesian coordinates is six: $x, y, z, \theta_x, \theta_y, \theta_z$

- **Infinite number of solutions:** if the number of joints in the robot (DOF) is more than the number of independent Cartesian coordinates that can be controlled within the robot's workspace. In this case, the robot would be a "redundant" robot.

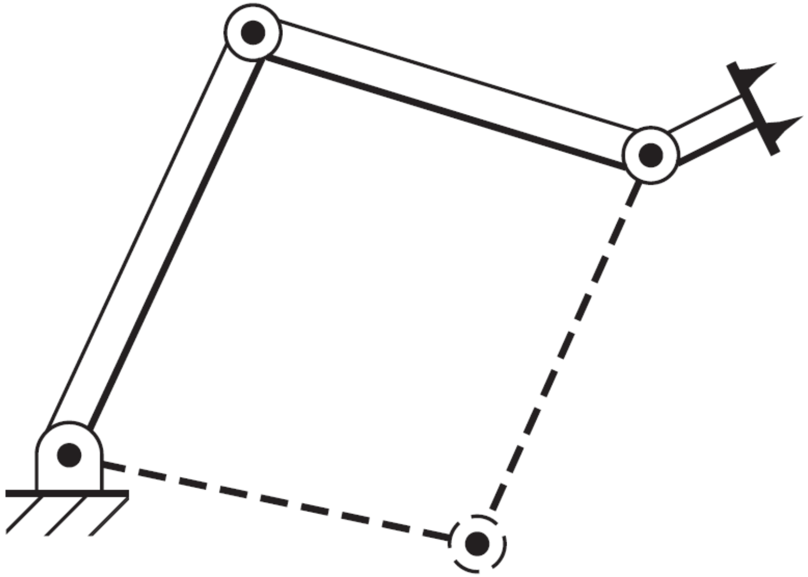
A **redundant robot is a manipulator that has **more degrees of freedom (DoF) than are required to perform a given task**.

Multiple Solutions

Example:

For the 3R planar manipulator, how many solutions can exist for reaching the same pose within the planar workspace of the robot? Give an example of how to choose one of the solutions.

The robot can have two different solutions (elbow up or elbow down).



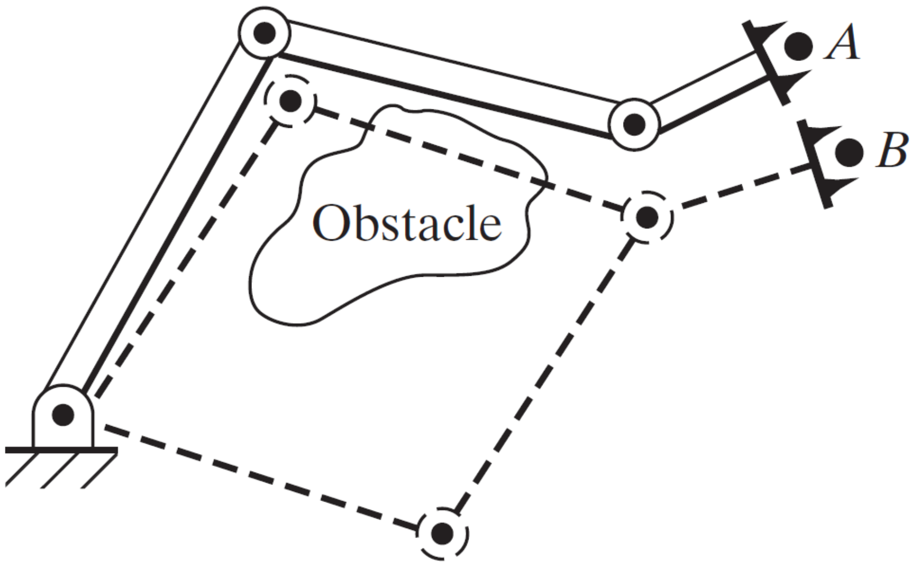
Multiple Solutions

Example:

For the 3R planar manipulator, how many solutions can exist for reaching the same pose within the planar workspace of the robot? Give an example of how to choose one of the solutions.

For example, to go from point A to point B when there is an obstacle, the robot can reach the target through two different configurations.

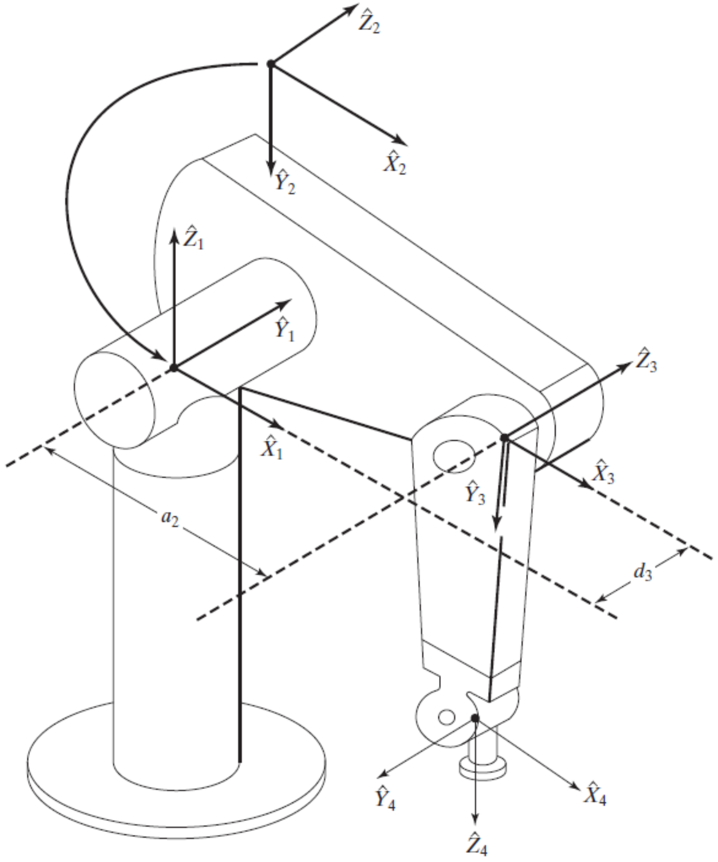
- The closest solution is when the elbow is up collision with the obstacle.
- The safest solution is when the elbow is down no collision, but farther motion.



Multiple Solutions

Example:

For the PUMA 560 robot, show four possible solutions for the same end-effector pose.

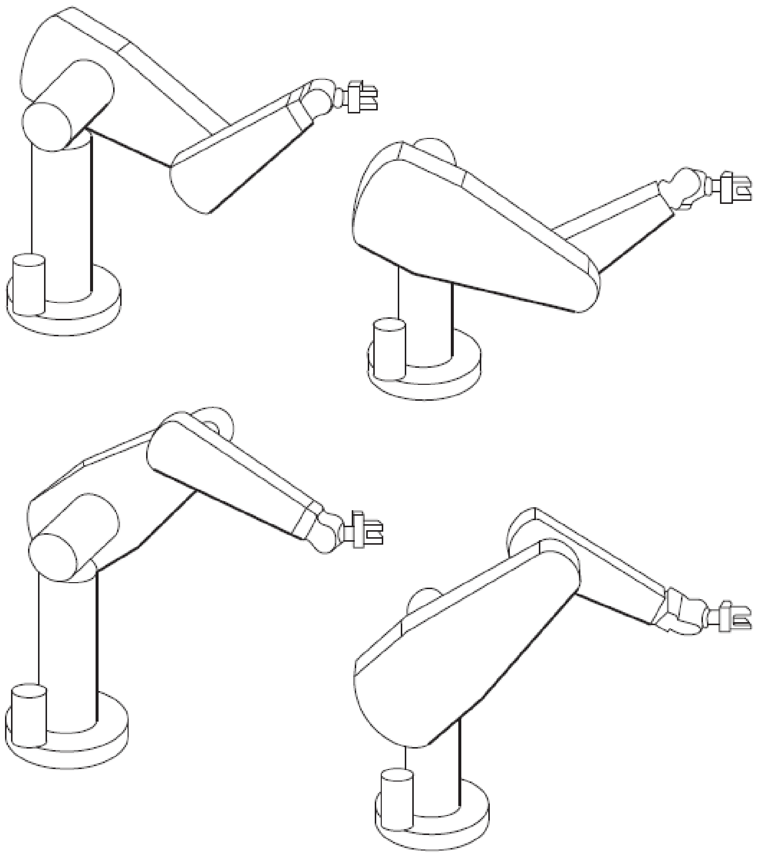


Multiple Solutions

Example:

For the PUMA 560 robot, show four possible solutions for the same end-effector pose.

- Top left upper arm outwards and elbow down.
- Top right upper arm inwards and elbow down.
- Lower left upper arm outwards and elbow up.
- Lower right - upper arm inwards and elbow up.

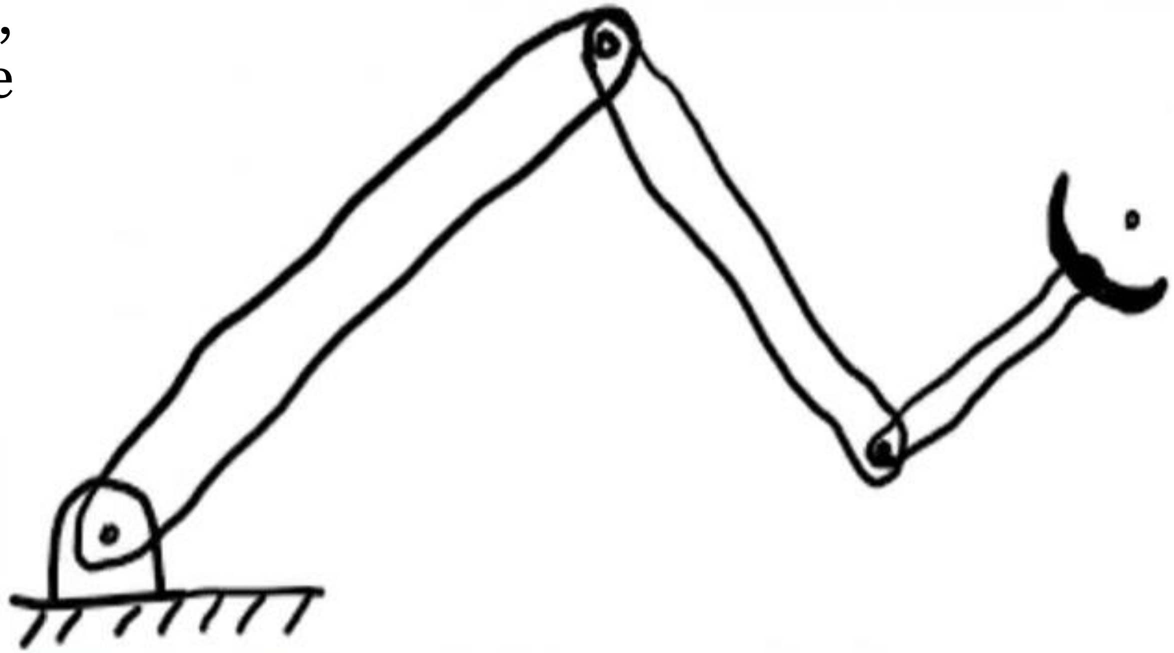


Multiple Solutions

Example:

For the 3R planar manipulator, how many solutions can exist for reaching the same point within the planar workspace of the robot (ignoring the orientation)?

Since the robot is 3-DOF and we are only controlling the end-effector's position (x, y), thus infinitely many solutions exist to reach the same (x, y) point within the workspace.



Inverse Kinematics Approaches

There are two types of approaches that can be used to find the solution of an inverse kinematics problem.

- Analytical Approach
  - Algebraic Approach
  - Geometric Approach

- Numerical Approach

Inverse Kinematics Approaches

- **Algebraic approach:**

Algebraic solutions can be achieved by comparing elements from the general transformation matrix to elements from the desired transformation matrix of the robot.

- **Geometric approach:**

Geometric solutions can be achieved by decomposing the spatial geometry of the robot into several plane-geometry problems. Joint angles can then be solved for by using the tools of plane-geometry.

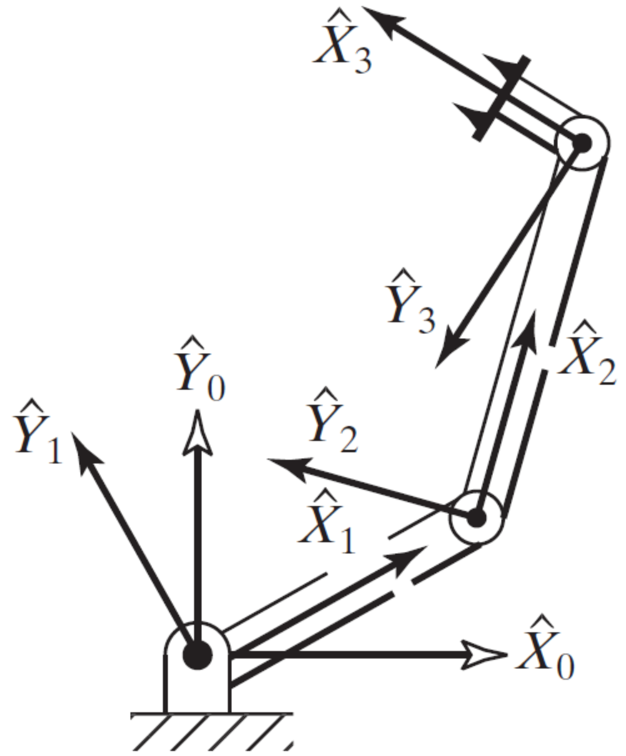
Algebraic Approach

Example:

For the 3R planar manipulator shown, the general transformation matrix that defines frame {3} relative to frame {0} is found through forward Kinematics as follows:

(a) Use inverse Kinematics to find the solution for the joint variables (in terms of x, y, $\emptyset$) for the robot to reach the following Cartesian pose at frame {3}.

(b) Then find the solution for the joint variables if $x = 3$ units, $y = 4$ units, $\emptyset = 150°$, $L1 = 2$ units, and $L2 = 5$ units.



$$
{}^0_3T = \begin{bmatrix} c_{123} & -s_{123} & 0.0 & l_1c_1 + l_2c_{12} \\ s_{123} & c_{123} & 0.0 & l_1s_1 + l_2s_{12} \\ 0.0 & 0.0 & 1.0 & 0.0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad {}^B_WT = \begin{bmatrix} c_\phi & -s_\phi & 0.0 & x \\ s_\phi & c_\phi & 0.0 & y \\ 0.0 & 0.0 & 1.0 & 0.0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

Algebraic Approach

Example:

(a) To find the three joint angles of the robot in terms of the given desired transformation matrix, we compare elements from the general transformation matrix and the desired transformation matrix.

$$c_\phi = c_{123},$$

$$s_\phi = s_{123},$$

$$x = l_1 c_1 + l_2 c_{12}$$

$$y = l_1 s_1 + l_2 s_{12}$$

$$x^2 + y^2 = l_1^2 + l_2^2 + 2 l_1 l_2 c_2$$

$$\sin(\theta_1 + \theta_2) = s_{12} = c_1 s_2 + s_1 c_2$$

$$\cos(\theta_1 + \theta_2) = c_{12} = c_1 c_2 - s_1 s_2$$

$$ {}^B_W T = \begin{bmatrix} c_\phi & -s_\phi & 0.0 & x \\ s_\phi & c_\phi & 0.0 & y \\ 0.0 & 0.0 & 1.0 & 0.0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^0_3 T = \begin{bmatrix} c_{123} & -s_{123} & 0.0 & l_1 c_1 + l_2 c_{12} \\ s_{123} & c_{123} & 0.0 & l_1 s_1 + l_2 s_{12} \\ 0.0 & 0.0 & 1.0 & 0.0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

Algebraic Approach

Example:

(a) For a solution to exist: $-1 < c_2 < 1$

$$\sin(\theta_1 + \theta_2) = s_{12} = c_1 s_2 + s_1 c_2$$
$$\cos(\theta_1 + \theta_2) = c_{12} = c_1 c_2 - s_1 s_2$$

$$c_2 = \frac{x^2 + y^2 - l_1^2 - l_2^2}{2 l_1 l_2}$$

$$x^2 + y^2 = l_1^2 + l_2^2 + 2 l_1 l_2 c_2$$

Algebraic Approach

Example:

(a) Use inverse Kinematics to find the solution for the joint variables (in terms of x, y, $\emptyset$) for the robot to reach the following Cartesian pose at frame {3}.

For a solution to exist: $-1 < c_2 < 1$

$$c_2 = \frac{x^2 + y^2 - l_1^2 - l_2^2}{2l_1 l_2}$$

$$s_2 = \pm\sqrt{1 - c_2^2}$$

$$\theta_2 = \text{Atan2}(s_2, c_2)$$

Algebraic Approach

Example:

(a) Use inverse Kinematics to find the solution for the joint variables (in terms of x, y, $\emptyset$) for the robot to reach the following Cartesian pose at frame {3}.

$$\theta_2 = \mathrm{Atan2}(s_2, c_2)$$

$$\theta_1 = A\tan 2(y, x) - A\tan 2\left(L_2 s_2, L_1 + L_2 c_2\right)$$

$$\theta_1 + \theta_2 + \theta_3 = \mathrm{Atan2}(s_\phi, c_\phi) = \phi$$

Algebraic Approach

Example:

For the 3R planar manipulator shown, the general transformation matrix that defines frame {3} relative to frame {0} is found through forward Kinematics as follows:

(b) To find the solution for the joint variables if $x = 3$ units, $y = 4$ units, $\emptyset = 150°$, $L1 = 2$ units, and $L2 = 5$ units, the desired transformation matrix would be.

$$ {}^{0}_{3}T_{desired} = \begin{bmatrix} -0.866 & -0.5 & 0 & 3 \\ 0.5 & -0.866 & 0 & 4 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ c_2 = \frac{3^2 + 4^2 - 2^2 - 5^2}{2(2)(5)} = -0.2 $$

$$ s_2 = \pm\sqrt{1-(-0.2)^2} = \pm 0.98 $$

Algebraic Approach

Example:

For the 3R planar manipulator shown, the general transformation matrix that defines frame {3} relative to frame {0} is found through forward Kinematics as follows:

(b) To find the solution for the joint variables if $x = 3$ units, $y = 4$ units, $\emptyset = 150°$, $L1 = 2$ units, and $L2 = 5$ units, the desired transformation matrix would be.

$$\boxed{\theta_2 = A\tan 2(\pm 0.98, -0.2) = 1.7721 \quad and \quad \text{-}1.7721 \quad radians}$$

$$\boxed{\theta_1 = A\tan 2(4,3) - A\tan 2\left(\pm 5(0.98), 2-5(0.2)\right) = \text{-}0.4422 \quad and \quad 2.2968 \quad radians}$$

$$\boxed{\theta_3 = A\tan 2(0.5, -0.866) + 0.4422 - 1.7721 = 1.2881 \quad radians}$$

$$\boxed{\theta_3 = A\tan 2(0.5, -0.866) - 2.2968 + 1.7721 = 2.0933 \quad radians}$$

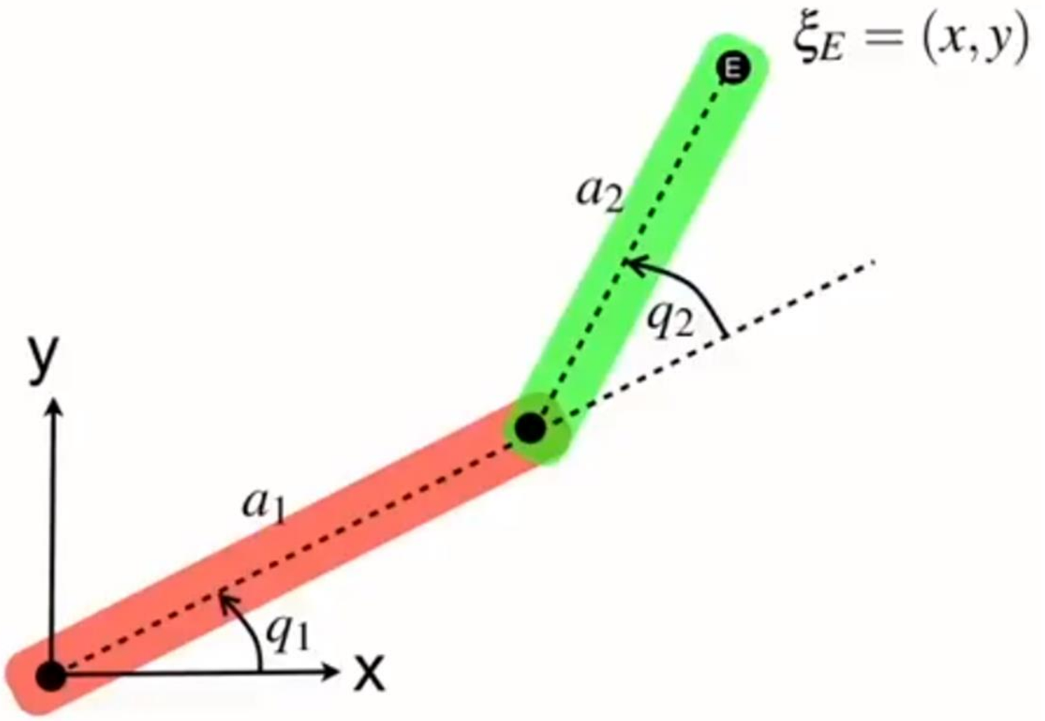
Outline

- Algebraic Approach
- **Geometric Approach**

Geometric Approach

Example:

For the shown 2R planar manipulator, determine the angles q_1 and q_2, given the coordinates x, y and the link lengths a_1 and a_2.



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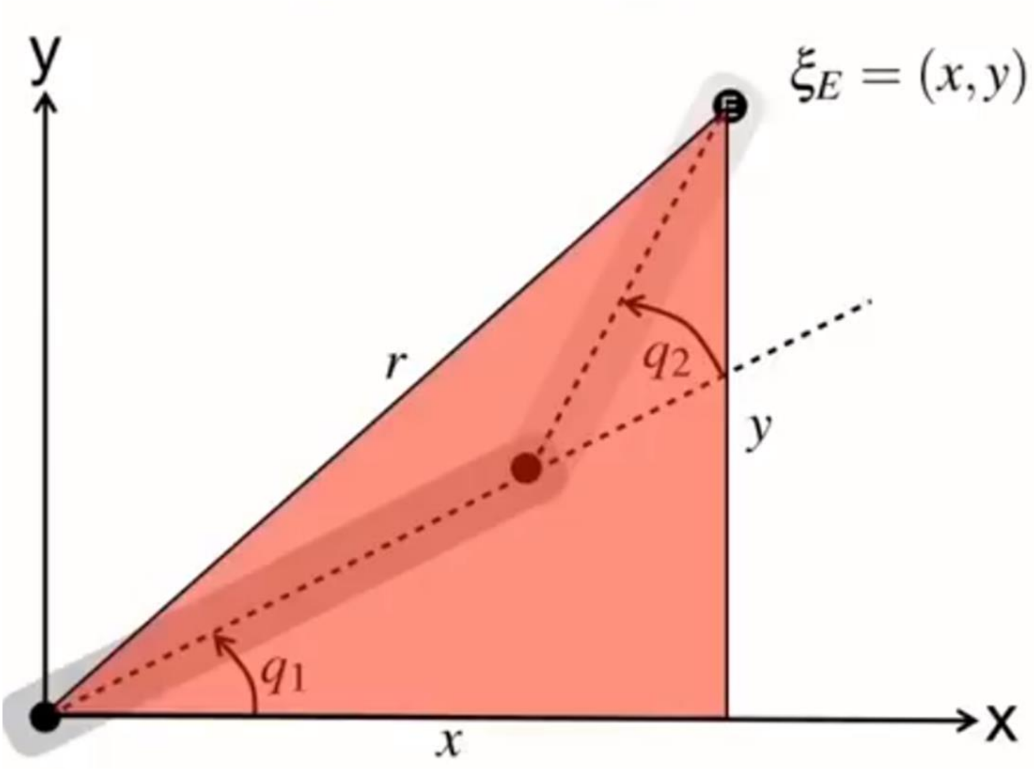
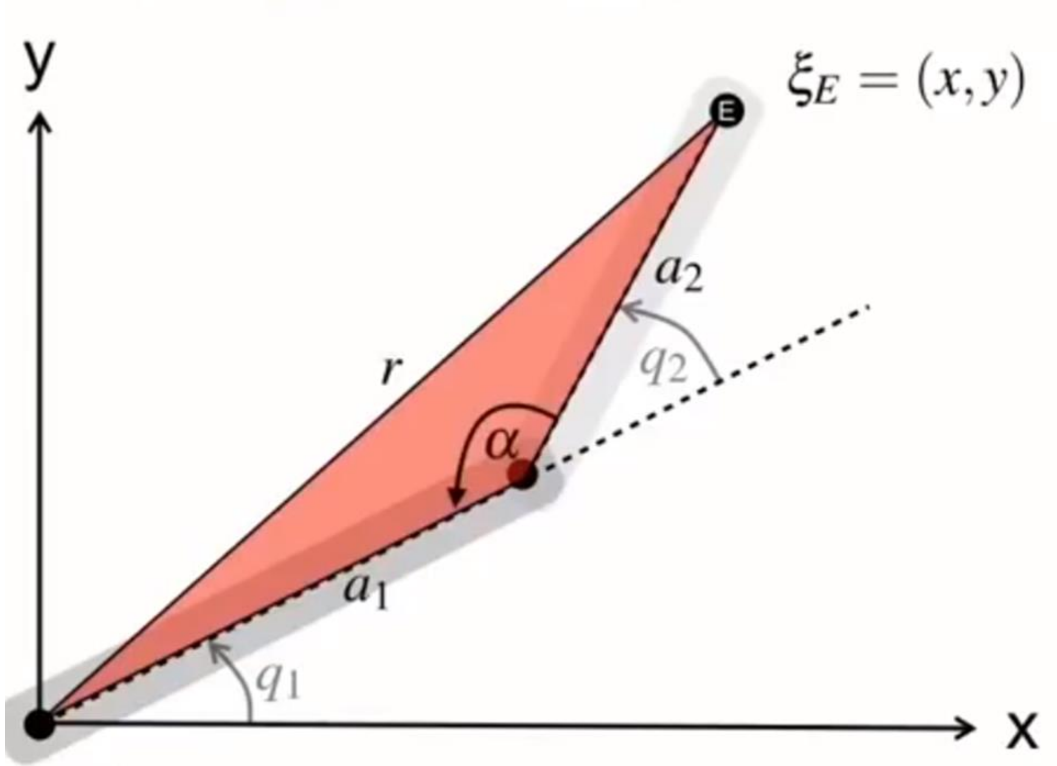


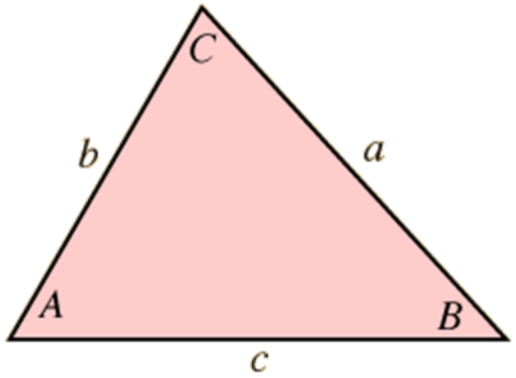
$$r^2 = x^2 + y^2$$

Example:

For the shown 2R planar manipulator, determine the angles q_1 and q_2, given the coordinates x, y and the link lengths a_1 and a_2.



$$c^2 = a^2 + b^2 - 2ab\cos C$$



Geometric Approach

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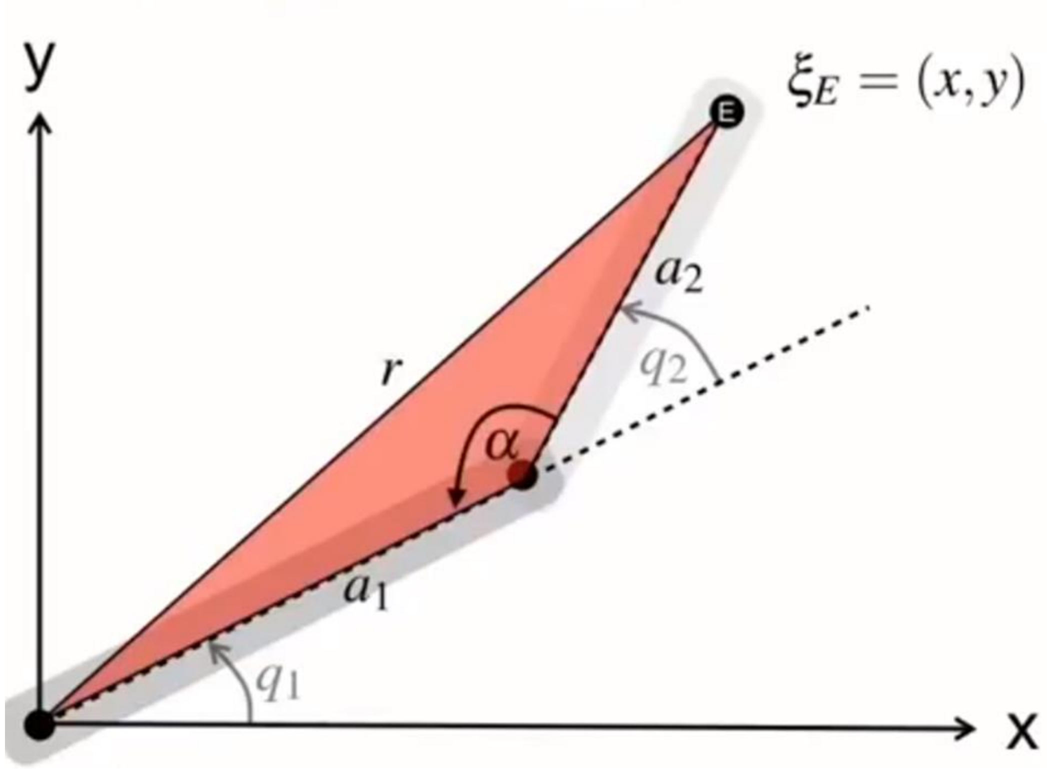


$$r^2 = x^2 + y^2$$

$$r^2 = a_1^2 + a_2^2 - 2a_1a_2\cos\alpha$$

$$\cos\alpha = \frac{a_1^2 + a_2^2 - r^2}{2a_1a_2}$$

$$= \frac{a_1^2 + a_2^2 - x^2 - y^2}{2a_1a_2}$$

$$q_2 = \pi - \alpha$$

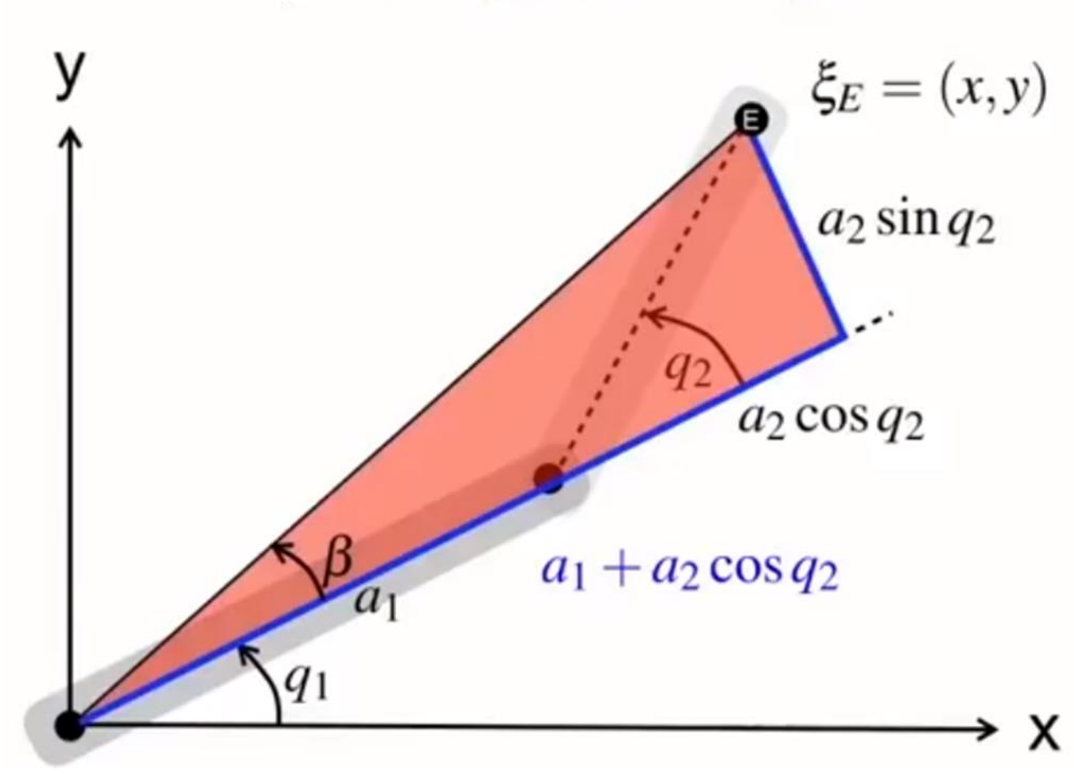
$$\cos q_2 = -\cos\alpha$$

$$\cos q_2 = \frac{x^2 + y^2 - a_1^2 - a_2^2}{2a_1a_2}$$

Geometric Approach

Example:

For the shown 2R planar manipulator, determine the angles q_1 and q_2, given the coordinates x, y and the link lengths a_1 and a_2.

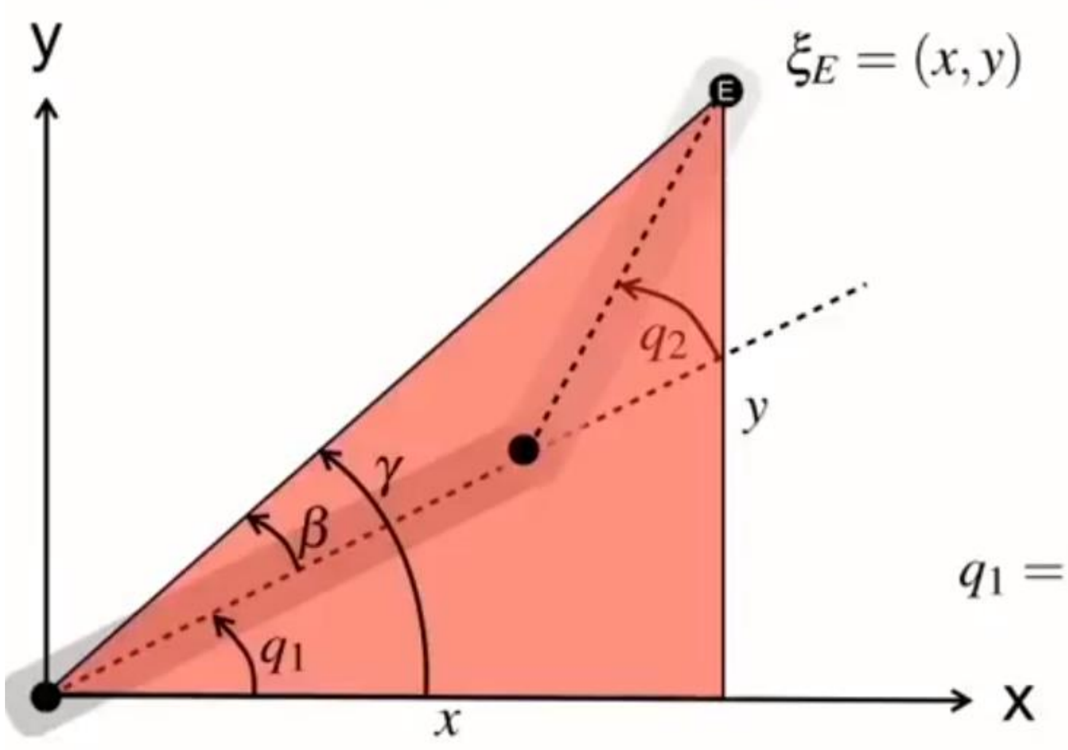


$$\beta = \tan^{-1} \frac{a_2 \sin q_2}{a_1 + a_2 \cos q_2}$$

Geometric Approach

Example:

For the shown 2R planar manipulator, determine the angles q_1 and q_2, given the coordinates x, y and the link lengths a_1 and a_2.



$$\gamma = \tan^{-1}\frac{y}{x}$$

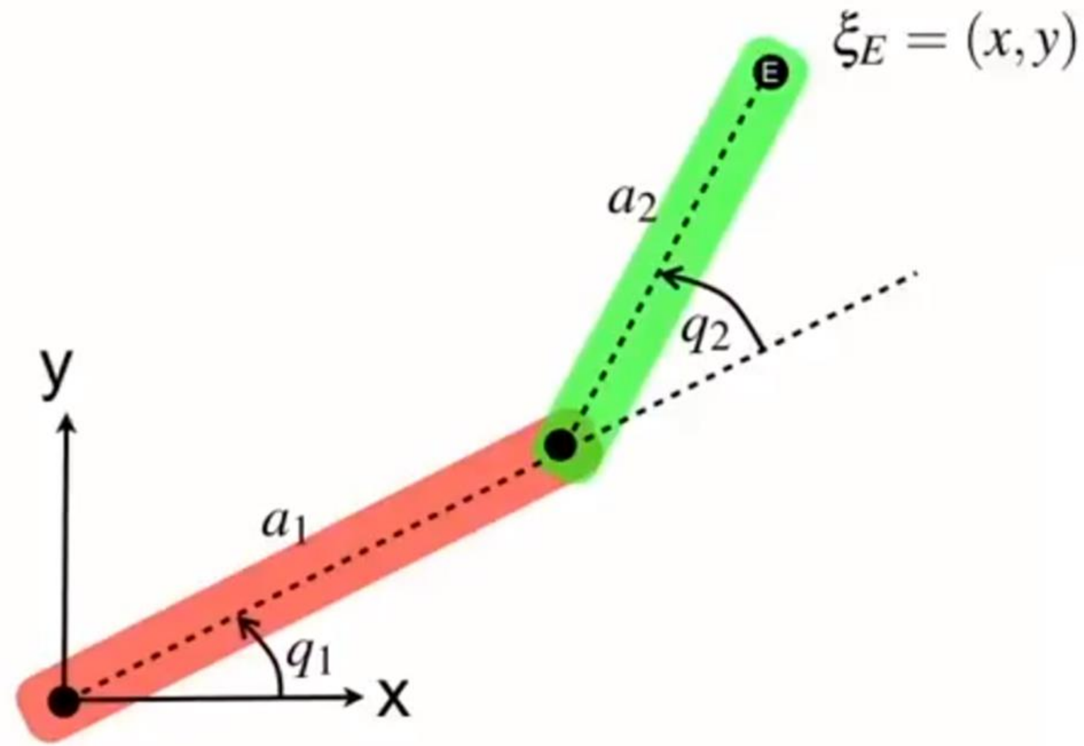
$$q_1 = \gamma - \beta$$

$$q_1 = \tan^{-1}\frac{y}{x} - \tan^{-1}\frac{a_2 \sin q_2}{a_1 + a_2 \cos q_2}$$

Geometric Approach

Example:

For the shown 2R planar manipulator, determine the angles q_1 and q_2, given the coordinates x, y and the link lengths a_1 and a_2.



$$\cos q_2 = \frac{x^2 + y^2 - a_1^2 - a_2^2}{2a_1a_2}$$

$$q_2 = \cos^{-1}\frac{x^2 + y^2 - a_1^2 - a_2^2}{2a_1a_2}$$

$$q_1 = \tan^{-1}\frac{y}{x} - \tan^{-1}\frac{a_2 \sin q_2}{a_1 + a_2 \cos q_2}$$

Geometric Approach

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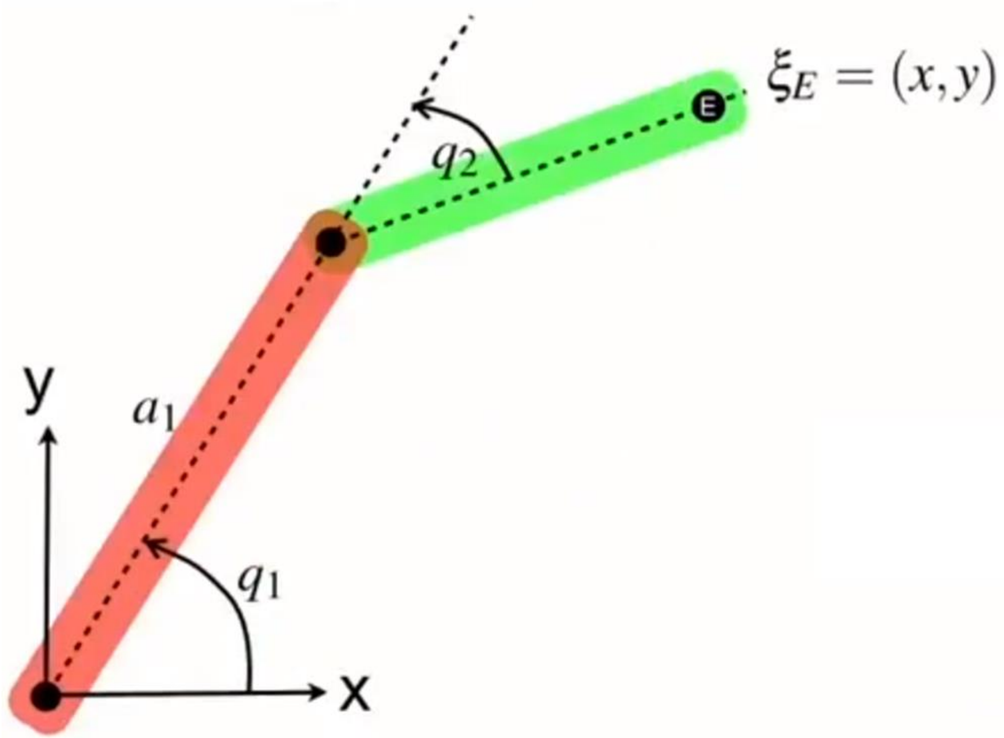
For the shown 2R planar manipulator, determine the angles q_1 and q_2, given the coordinates x, y and the link lengths a_1 and a_2.



$$q_2 = -\cos^{-1}\frac{x^2+y^2-a_1^2-a_2^2}{2a_1a_2}$$

Geometric Approach

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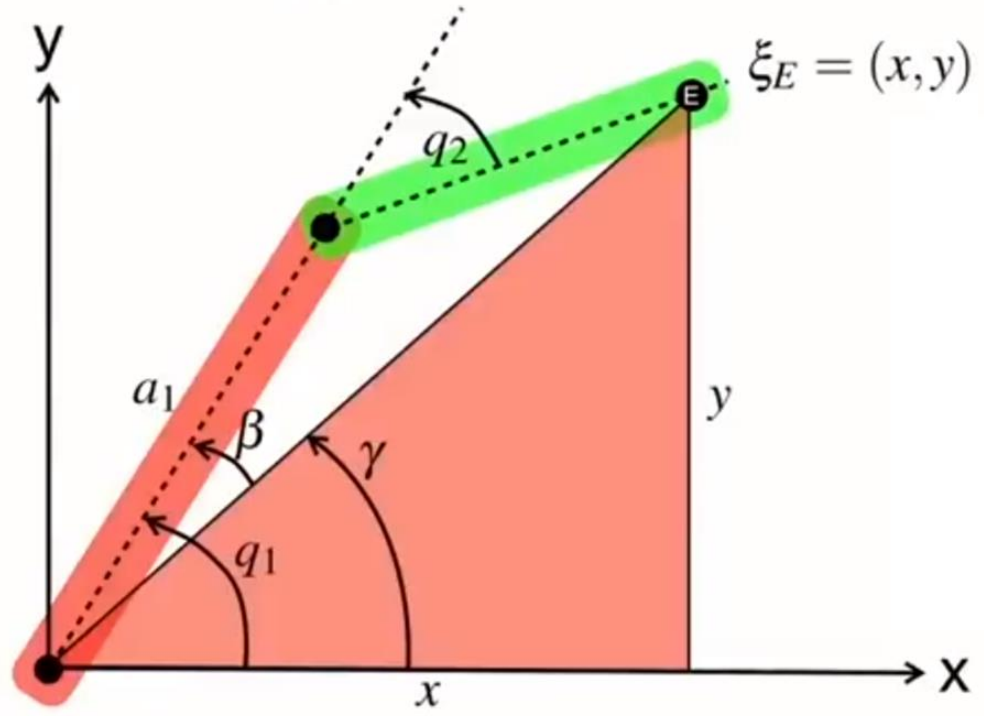

$$\gamma = \tan^{-1}\frac{y}{x}$$

$$q_1 = \gamma + \beta$$

$$q_1 = \tan^{-1}\frac{y}{x} + \tan^{-1}\frac{a_2 \sin q_2}{a_1 + a_2 \cos q_2}$$

Geometric Approach

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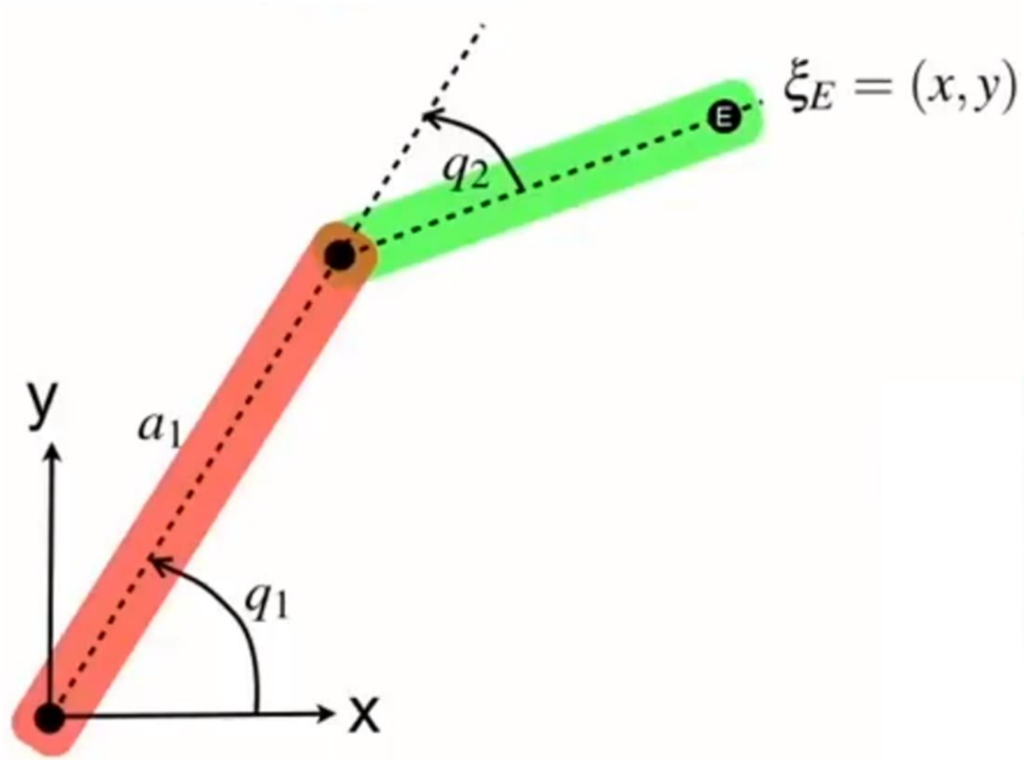
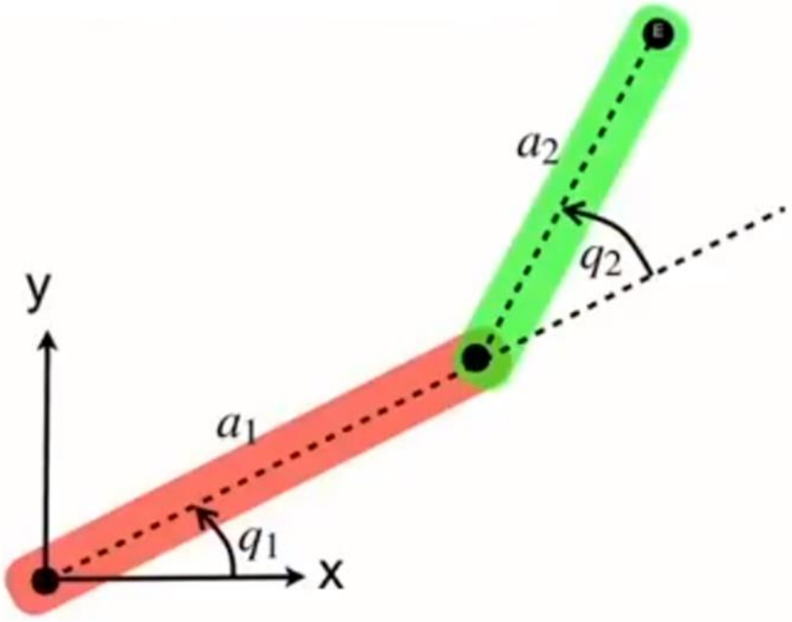
For the shown 2R planar manipulator, determine the angles q_1 and q_2, given the coordinates x, y and the link lengths a_1 and a_2.

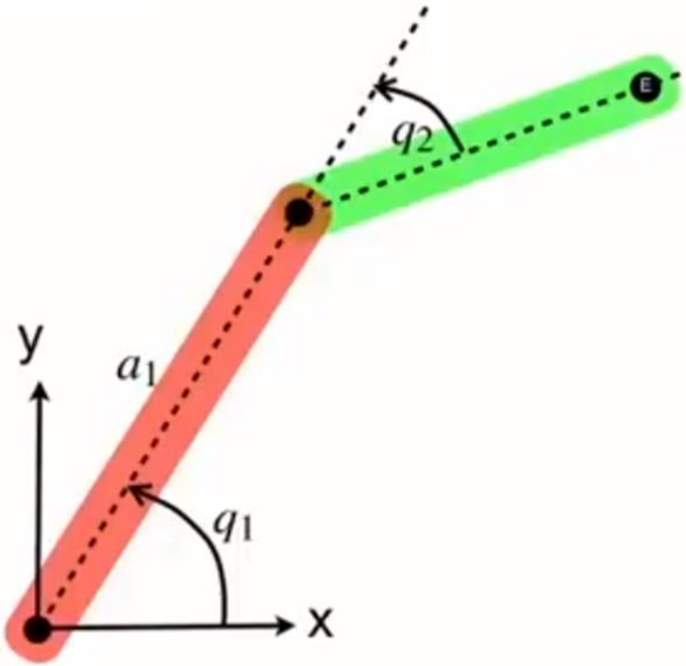


$$\boxed{q_2 = -\cos^{-1}\frac{x^2+y^2-a_1^2-a_2^2}{2a_1a_2}}$$

$$\boxed{q_1 = \tan^{-1}\frac{y}{x} + \tan^{-1}\frac{a_2\sin q_2}{a_1 + a_2\cos q_2}}$$

Geometric Approach





$$q_2 = \cos^{-1} \frac{x^2 + y^2 - a_1^2 - a_2^2}{2a_1a_2}$$

$$q_1 = \tan^{-1} \frac{y}{x} - \tan^{-1} \frac{a_2 \sin q_2}{a_1 + a_2 \cos q_2}$$

$$q_2 = -\cos^{-1} \frac{x^2 + y^2 - a_1^2 - a_2^2}{2a_1a_2}$$

$$q_1 = \tan^{-1} \frac{y}{x} + \tan^{-1} \frac{a_2 \sin q_2}{a_1 + a_2 \cos q_2}$$

Any Questions?